

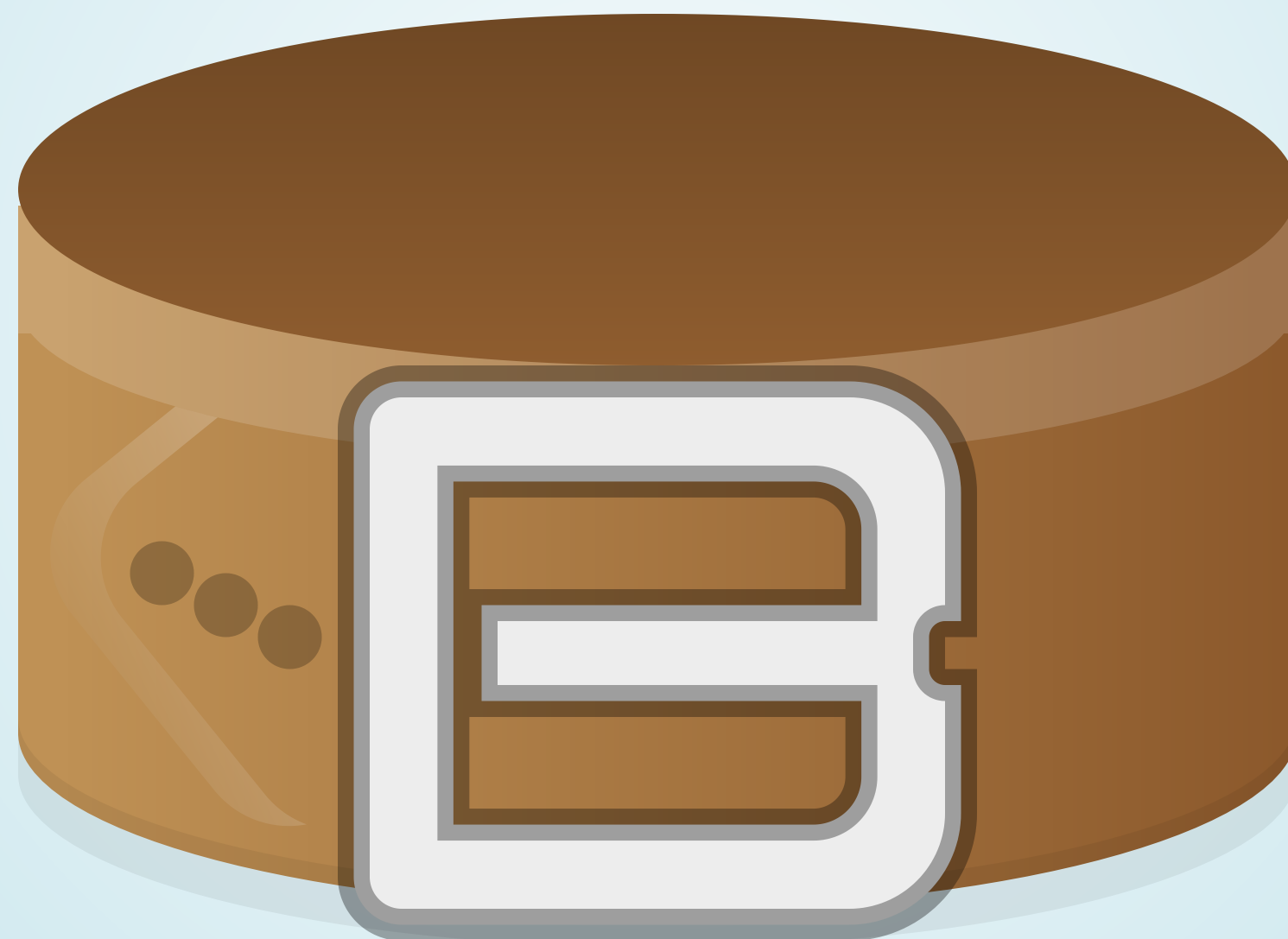
Practical ReasonML



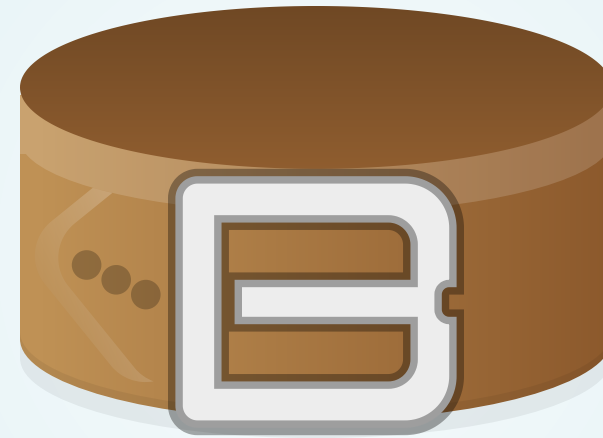
@MarcoEmrich

BS

BUCKLESCRIPT



REASONML / OCAML

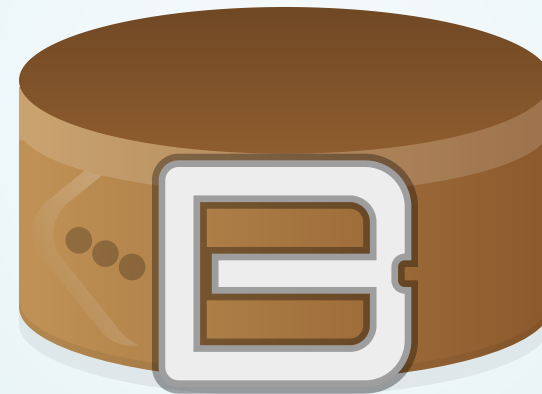


JAVASCRIPT

REASONML / OCAML



BSB-NATIVE



NATIVE / BACKEND / MICRO KERNEL / REACT NATIVE

TRANSPILE TO JS



BLOOMBERG



- 176 Offices
- 19000 Employees

HELLO WORLD

```
npm -g i bs-plattform  
bsb -init sample -theme basic-reason
```

DEMO

COMPUTER OR HUMAN?

stolen from Sean Grove

WRITTEN BY A MACHINE? (1/2)

```
var Int_map = require("./int_map.js");
function test() {
  var m = /* Empty */0;
  for(var i = 0; i <= 10000000; ++i) {
    m = add(i, i, m);
  }
  for(var j = 0; j <= 10000000; ++j) {
    find(j, m);
  }
  return /* () */0;
}
```


WRITTEN BY A MACHINE? (2/2)

```
var Pervasives = require("bs-plattform/lib/js/pervasives");
var Http       = require("http");

var hostname = "127.0.0.1";

function create_server(http) {
  var server = http.createServer(function (_, resp) {
    resp.statusCode = 200;
    resp.setHeader("Content-Type", "text/plain");
    return res.end("Hello World\n");
  });
  return server.listen(3000, hostname, function () {
    console.log("Server running at http://") +
    . . .
```

SOLUTION

GENERATES
IDOMATIC JS

REASON

?

TypeScript



DEADLINES?

TIME PRESSURE?

ENEMIES OF THE TYPE-SYSTEM

```
: any
```

```
@ts-ignore
```

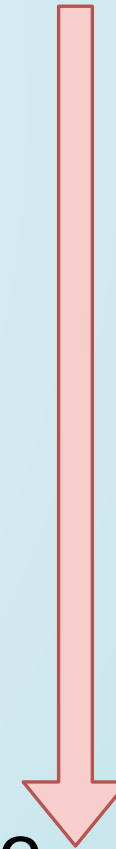
AnyScript™

JS-TRANSPILER LANGUAGES

Purity Enforcement /
Type Coverage



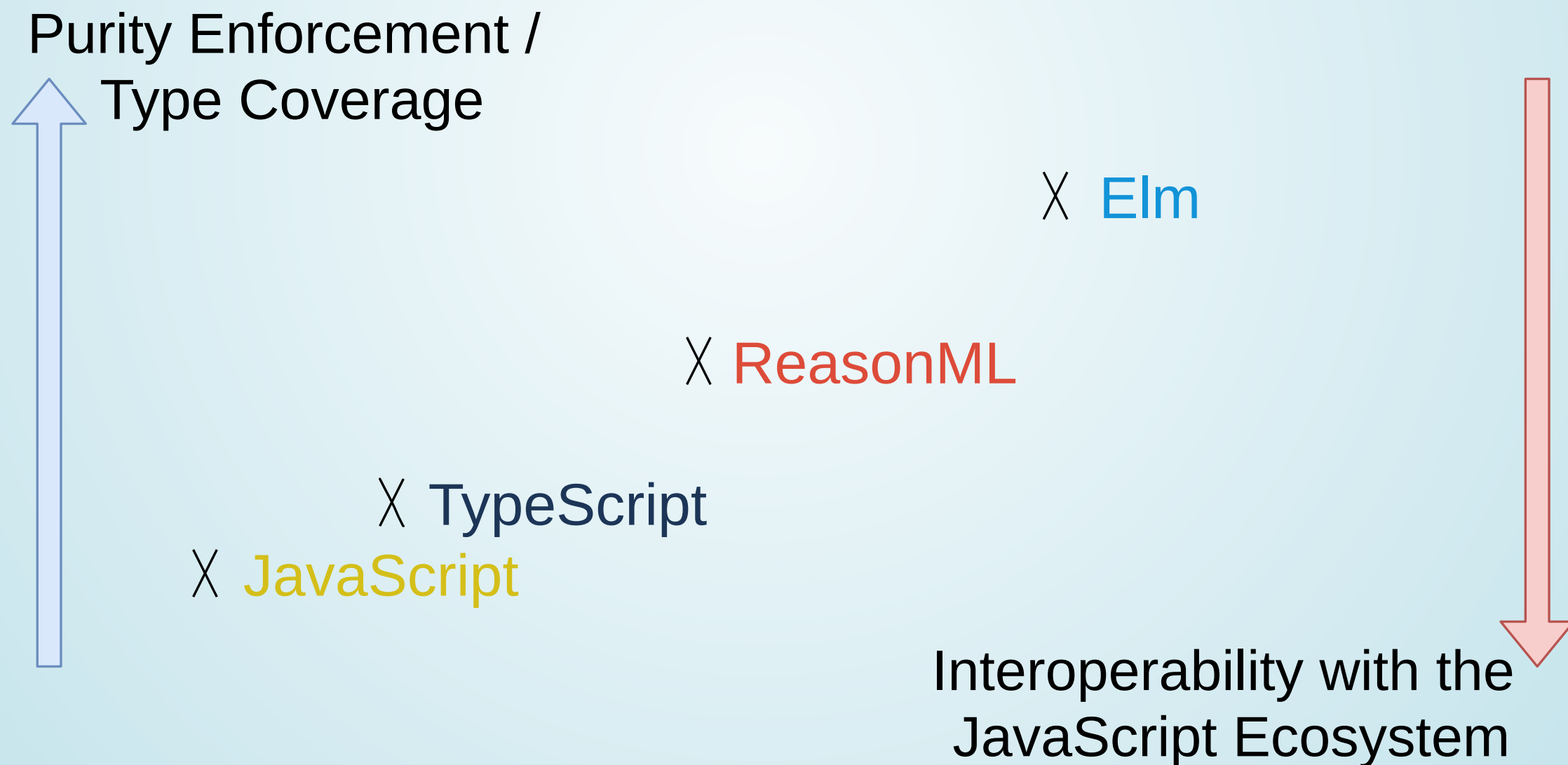
Interoperability with the
JavaScript Ecosystem



ELM



JS-TRANSPILER LANGUAGES



CHOOSE
BEST TOOL
FOR THE JOB

Need a lot of crazy JS-Libraries?

Need no JS-Libraries at all?

Somewhere in between?

REASON

THREE WISHES





BENEFIT OF FUNCTIONAL PROGRAMMING

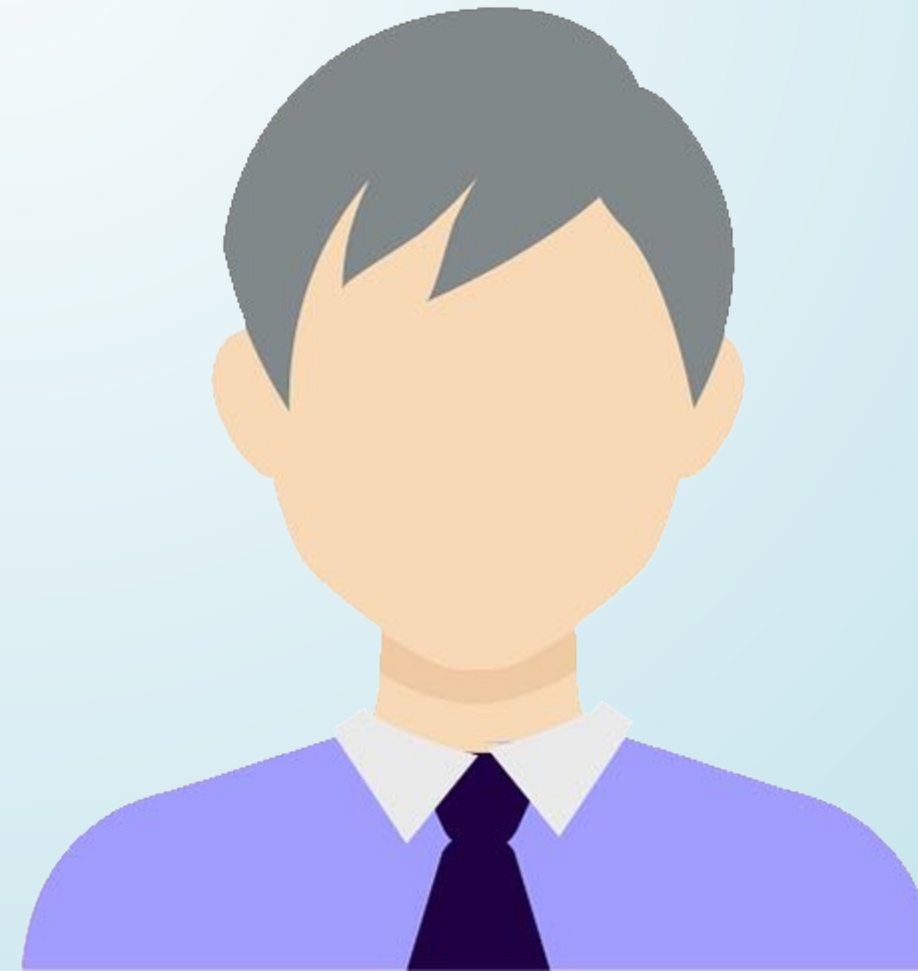
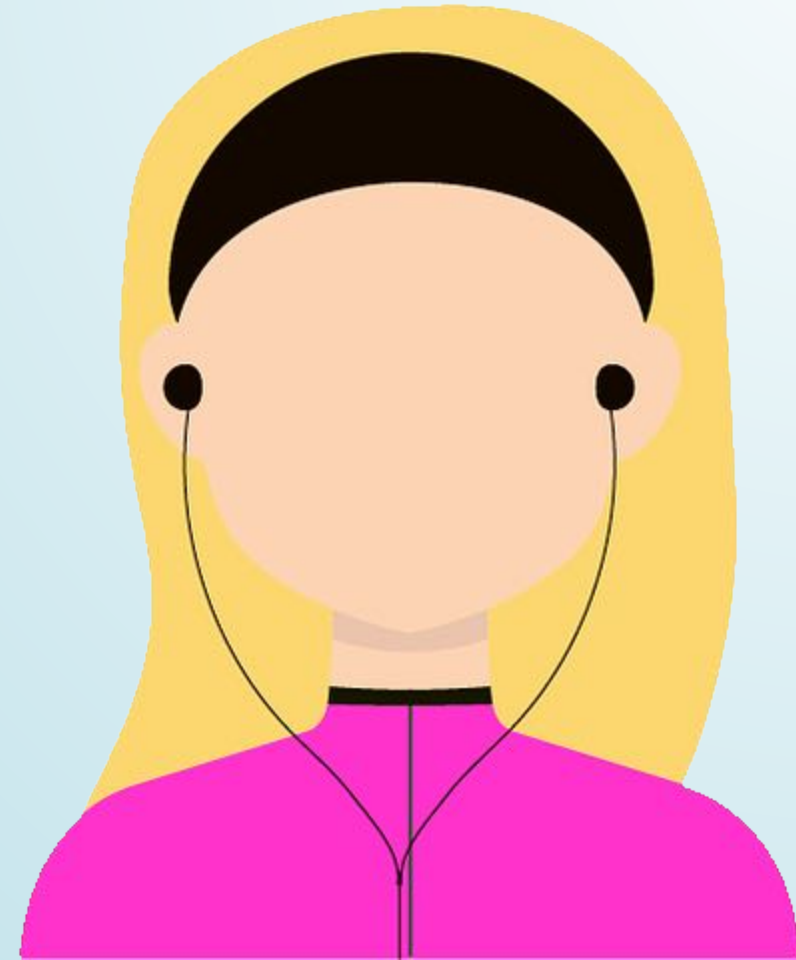
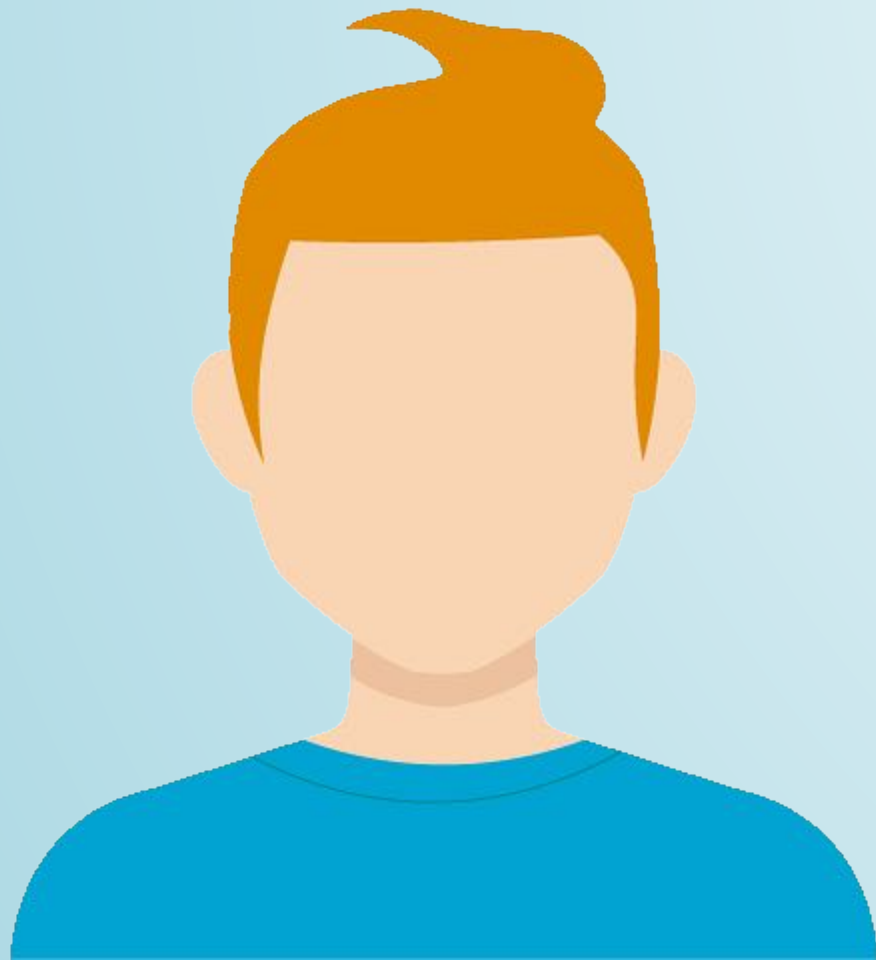
... (IN THE FRONT END)

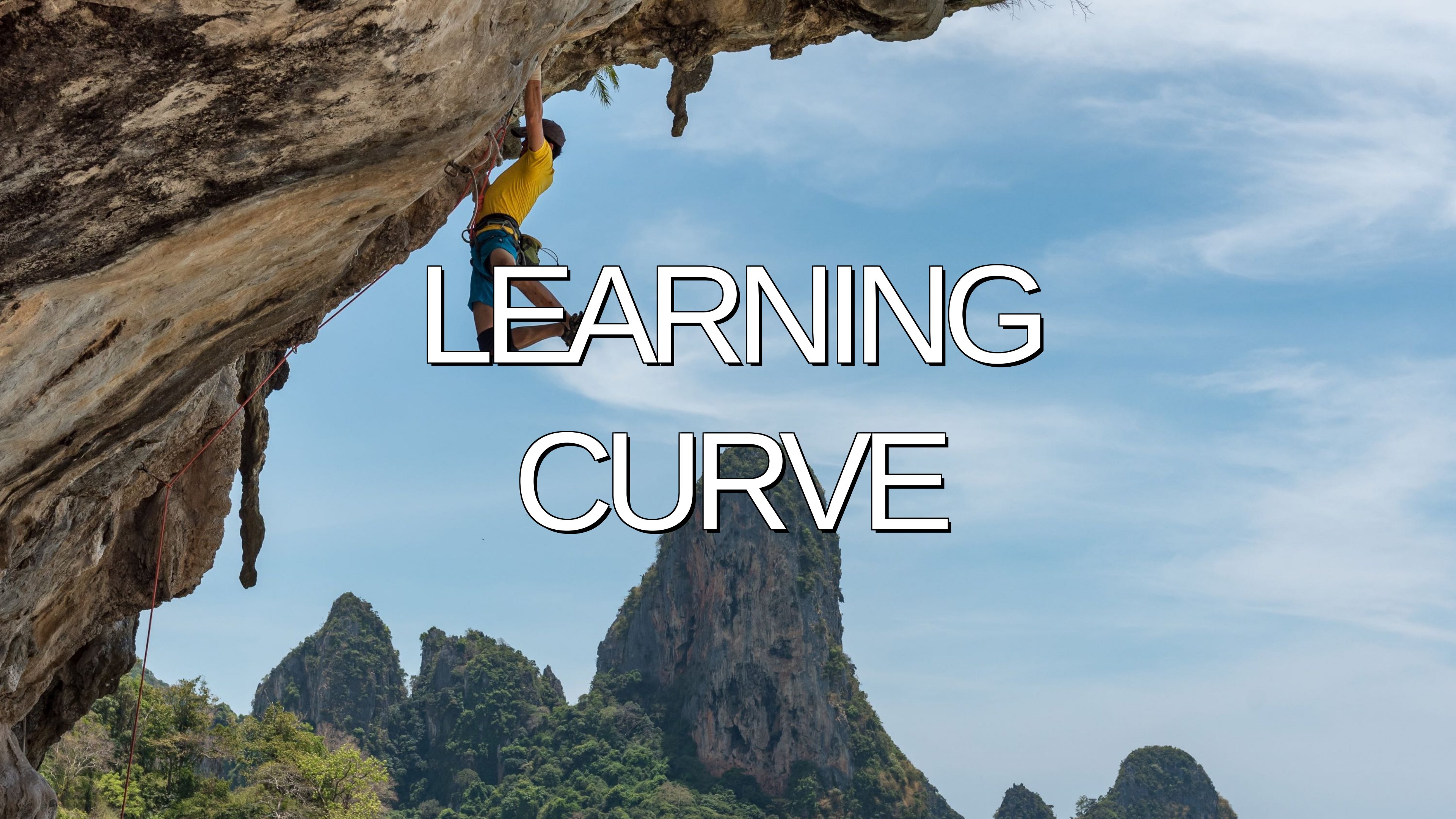


PURESRIPT, ELM,
HASKELL ...

BUT...

... CONVINCE COLLEAGUES



A person in a yellow shirt and blue shorts is climbing a steep, overhanging rock face. A red rope is attached to the rock and runs down the side. The background features a blue sky with light clouds and several jagged, forested rock peaks in the distance.

LEARNING CURVE

CURSE OF THE MONAD

Zygohistomorphic prepromorphisms



Used when you really need both semi-mutual recursion and history and to repeatedly apply a natural transformation as you get deeper into the functor. Zygo implements semi-mutual recursion like a zygomorphism. Para gives you access to your result à la paramorphism.

```
import Control.Morphism.Zygo
import Control.Morphism.Prepro
import Control.Morphism.Histo
import Control.Functor.Algebra
import Control.Functor.Extras

zygoHistoPrepro
  :: (Unfoldable t, Foldable t)
  => (Base t b -> b)
  -> (forall c. Base t c -> Base t c)
  -> (Base t (EnvT b (Stream (Base t)) a) -> a)
  -> t
  -> a
zygoHistoPrepro f g t = gprepro (distZygoT f distHisto) g t
-- unless you want a generalized zygomorphism.
```

FP

EASY TO LEARN



VERY, VERY, VERY, VERY,
VERY

AWESOME

TYPE SYSTEM



SHOVEL S**T





REASON... WISHES BECOME TRUE



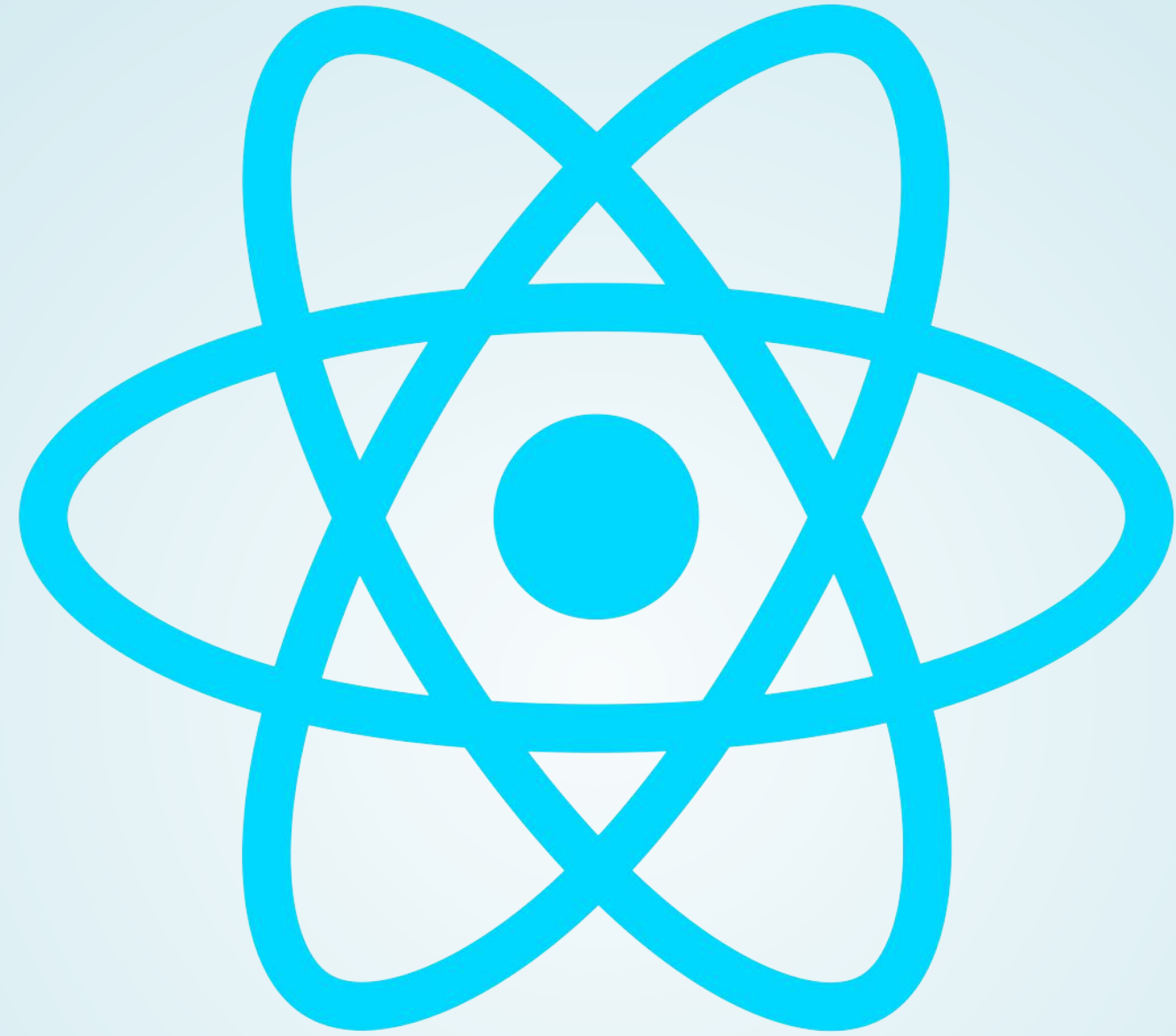
1. Functional Programming
2. Awesome Type System
3. Ability to Shovel S**t

HISTORY



JORDAN WALKE

@ JSConfUS 2013



REACTJS



OCaml is an industrial strength programming language supporting functional, imperative and object-oriented styles





How about OCAML in the Frontend?

WAT?

WTF?

*yan batlhchaj semi-mutual recursion
'ej qun qar blmejnIS 'ej natural
transformation qeSmeyllj Suq deeper
vaj functor. semi-mutual recursion rur
zygomorphism implements zygo.
naw' SoH nob para ghot'e' à De la
paramorphism.*

— zygohistomorphic prepromorphisms
(official Haskell Documentation)

[Klingon Translation]

**SYNTAX
MATTERS!**

REASON

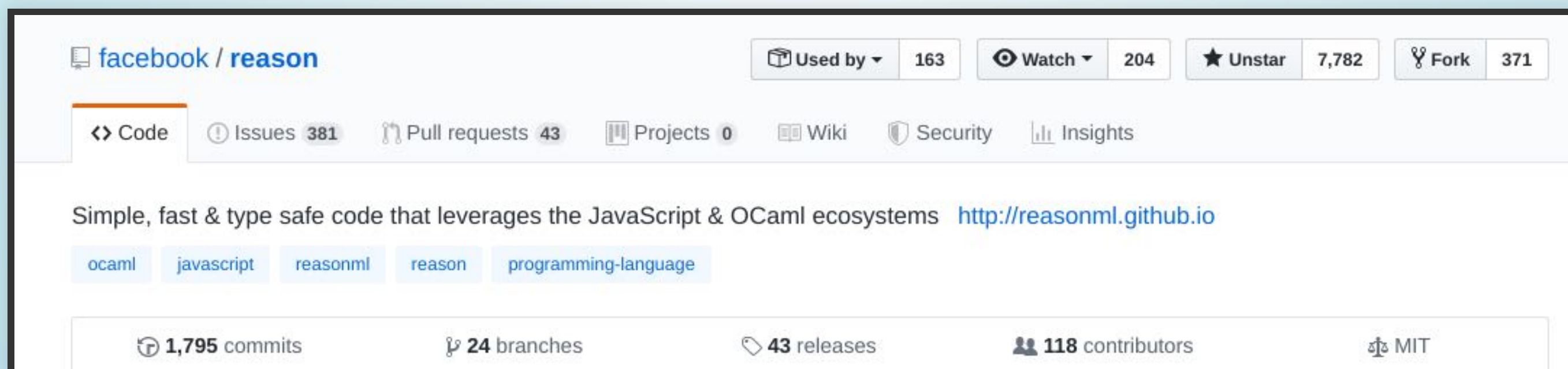
REASONML

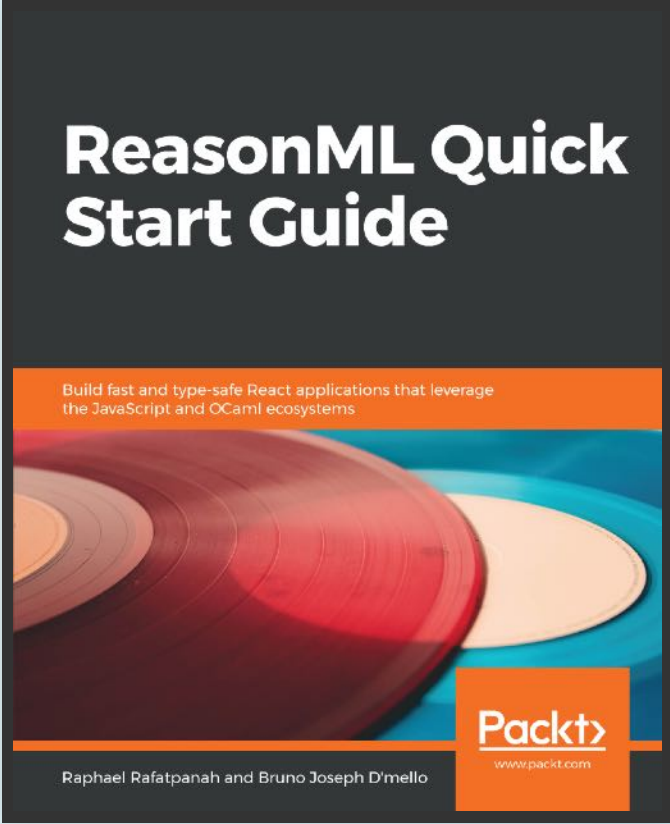
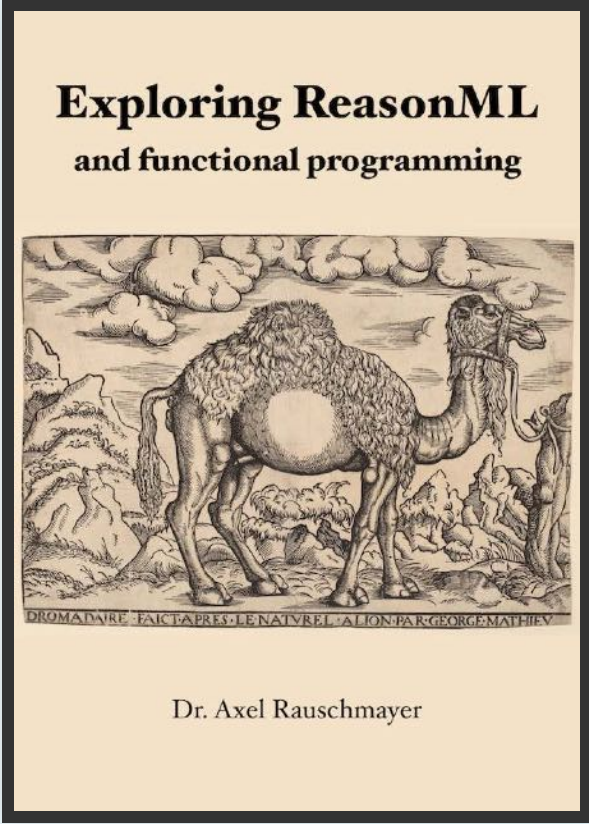
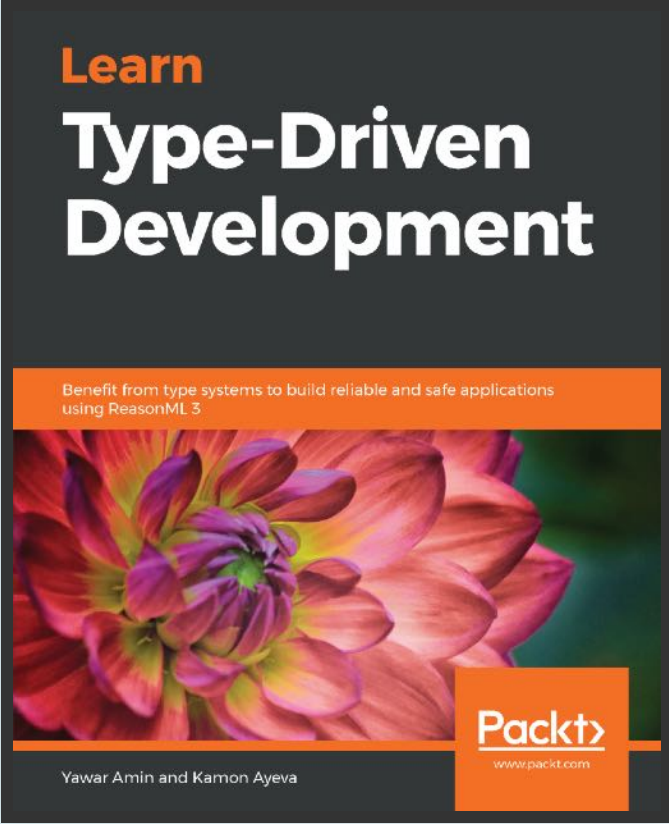
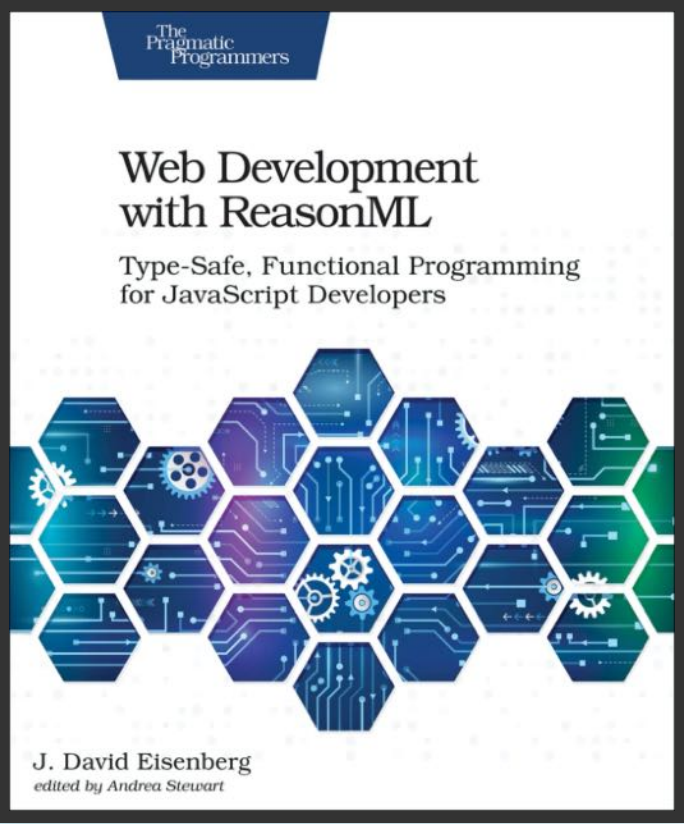
=

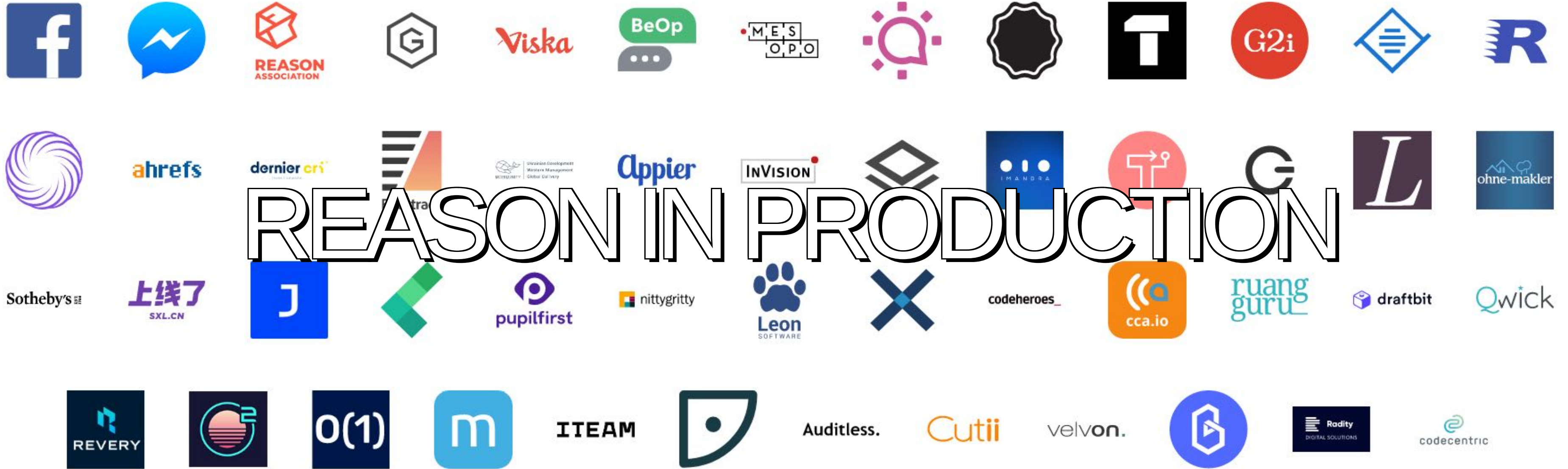
OCAML + JS-LIKE SYNTAX

STATISTICS

- > 120 Contributors
- > 8500 Stars on Github
- Big Companies
- 25 Years of OCAML











STRING CALCULATOR KATA



BY ROY OSHEROVE

"1,2,3" $\Rightarrow$ 6

NEW REQUIREMENT

IGNORE $\geq$ 1000

"1,1000,2" $\Rightarrow$ 3


```
let splitByComma = str => Js.String.split(",", str);  
let mapToInt = numbers => List.map(int_of_string, numbers);  
let lessThan1000 = numbers => List.filter(n => n < 1000, numbers)  
let sum = numbers => List.fold_left((+), 0, numbers);
```

AUTO-CURRYING

```
let splitByComma = str => Js.String.split(",", str);  
let mapToInt = numbers => List.map(int_of_string, numbers);  
let lessThan1000 = numbers => List.filter(n => n < 1000, numbers)  
let sum = numbers => List.fold_left((+), 0, numbers);
```



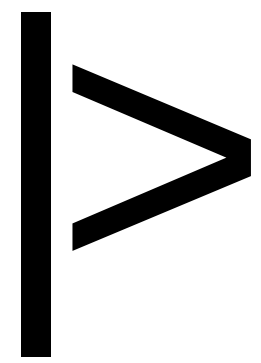
```
let splitByComma = Js.String.split(",");  
let mapToInt = List.map(int_of_string);  
let lessThan1000 = List.filter(n => n < 1000);  
let sum = List.fold_left((+), 0);
```

COMPOSITION

```
let stringCalc = str =>  
  sum(lessThan1000(mapToInt(splitByComma(str))));
```

(((((WTF???))))

PIPELINE



```
let stringCalc = str => str  
  |> splitByComma  
  |> mapToInt  
  |> lessThan1000  
  |> sum;
```

POINT FREE

```
open Rationale.Function;
```

```
let stringCalc =  
  ||> splitByComma  
  ||> mapToInt  
  ||> lessThan1000  
  ||> sum;
```



WAR OF TYPES

STATIC

VS

DYNAMIC

TYPE SYSTEMS

- Benefits
- Cost

COST

- Noise
- Maintenance
- Complexity

BENEFITS

- Type Safety
- Documentation
- IDE-Convenience
- Optimization

BETTER RATIO

- increase Benefits ↑
- reduce Cost ↓

REDUCE COSTS ↓

- Type Inference
- Structural Typing

INCREASE BENEFITS ↑

- Tagged Types
- Variant Types



BILLION DOLLAR
MISTAKE

NULL

Tony Hoare - 1964



<https://www.infoq.com/presentations/Null-References-The-Billion-Dollar-Mistake-Tony-Hoare>

REASONML

?

REASONML

pull

REASONML

none

OPTION

```
type option('a) = None | Some('a);
```

REASON TYPE SYSTEM

- Types can be inferred
- Coverage is always 100%.
- No need for a "type coverage" tool
- completely "sound"

IDE-SUPPORT

- Language Server
- VS-Code
- Atom
- vim
- Emacs
- Idea / IntelliJ / Webstorm
- Sublime



SHOVEL S**T



SHOVEL S**T

- Easy integration with JS
- Mutability if Needed
- High Performance

JAVASCRIPT INTEROP

```
Js.log("this is reason");
```

```
[%bs.raw {| console.log('here is some javascript for you') |}];
```

```
let y = [%bs.raw {| 'something' |}];  
Js.log(("a string" ++ y, 10 + y));
```



```
let y: string = [%bs.raw {| 'well-typed' |}];  
Js.log(("a string" ++ y, 10 + y));
```



```
let jsCalculate: (array(int), int) => int = [%bs.raw
{|
  function (numbers, scaleFactor) {
    var result = 0;
    numbers.forEach(number => {
      result += number;
    });
    return result * scaleFactor;
  }
}];

let calculate = (numbers, scaleFactor) =>
  jsCalculate(Array.of_list(numbers), scaleFactor);

Js.log(calculate([1, 2, 3], 10)); /* -> 60 */
```

EMBED A JSX-COMPONENT

```
[@bs.module "./LoadingNotification"] [@react.component]  
external make: (~unknownCount: int, ~unclearCount: int) =>  
  React.element =  
    "LoadingNotification";
```

```
<LoadingNotification unkownCount=3 unclearCount=6>  
... 
```

REASON PACKAGE INDEX

🔍 Search packages

react 61 utilities 42 ui 29 boilerplate 22 testing 16 development tools 15 async 10 graphics 10 platform api 10 css 9
database 9 graphql 8 cli 7 collections 7 data fetching 7 data serialization 7 cloud service api 6 json 6 opengl 6
react-native 6 code generation 5 routing 5 sql 5 date/time manipulation 4 dom 4 http client 4 parsing 4
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state management 3 express 2 filesystem 2 game development 2 http server 2 reactive programming 2 ssr 2
string manipulation 2 svg 2 configuration 1 error tracking 1 geolocation 1 i18n 1 maps 1 nosql 1 presentation 1
regular expressions 1 web framework 1 xml 1

RECENT RELEASES

bs-priority-queue	0.1.4	5 days ago
bs-ws	1.0.9	5 days ago
reductive-dev-tools	0.1.4	5 days ago
rationale	0.1.10	5 days ago
@mobily/re-date	0.8.2	6 days ago

MOST POPULAR

reason-react	0.5.3	2071 ★
@glennsl/bs-json	3.0.0	168 ★
reason-apollo	0.15.1	398 ★
bs-react-native	0.10.0	445 ★
@phenomic/reason	1.0.0-alpha.13	3133 ★

ESCAPE-
HATCHES

PURITY

GETTING S**T DONE

PURE BY DEFAULT

- Immutable by Default
- Compile before Runtime Errors

BUT



```
let counter = ref(5);  
counter := counter^ + 1  
Js.log(counter) // => 6
```

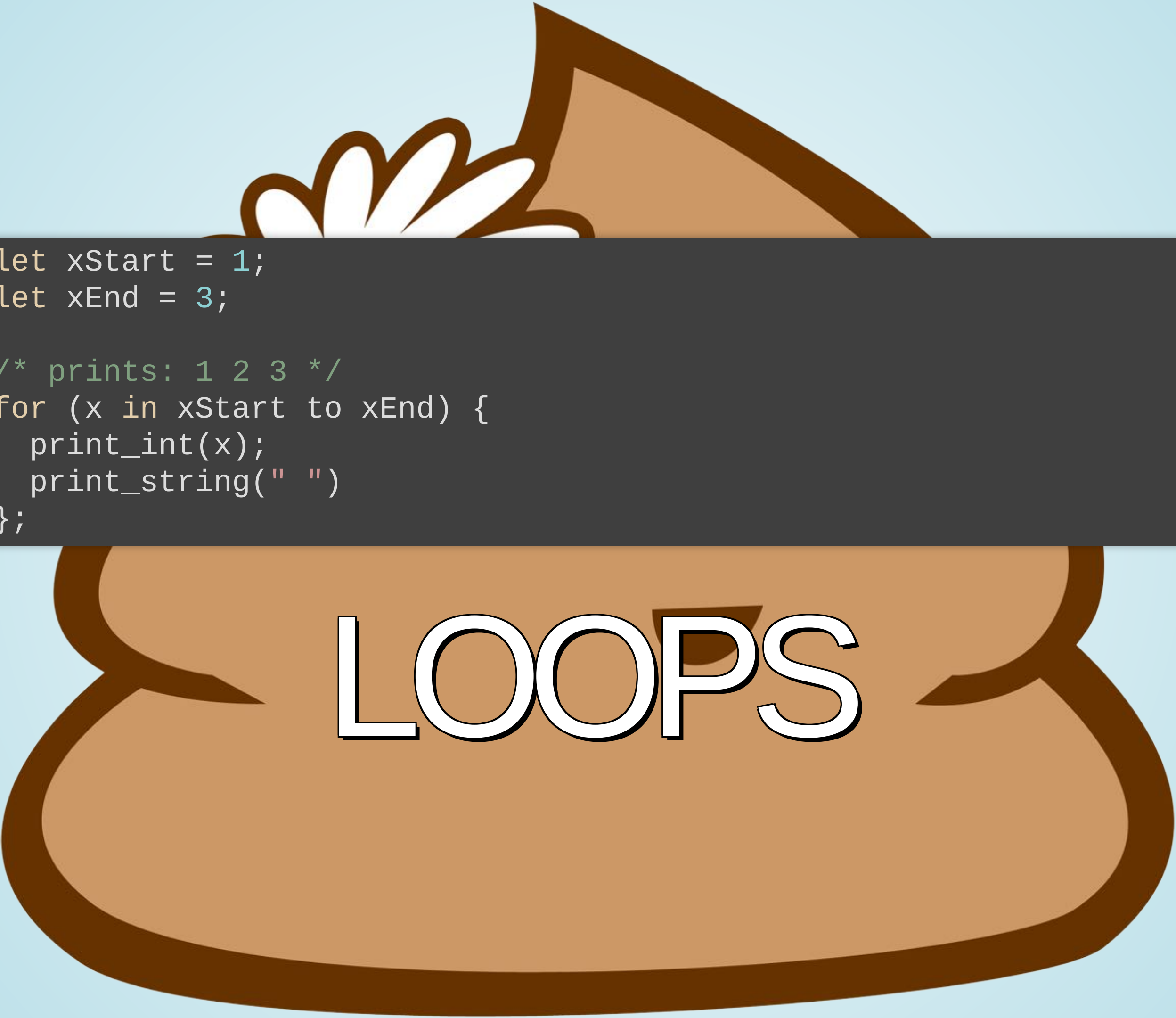
MUTATION



```
let getItem = (theList) =>
  if (callSomeFunctionThatThrows()) {
    /* return the found item here */
  } else {
    raise(Not_found)
  };

let result =
  try (getItem([1, 2, 3])) {
    | Not_found => 0 /* Default value if getItem throws */
  };
```

EXCEPTIONS



```
let xStart = 1;  
let xEnd = 3;  
  
/* prints: 1 2 3 */  
for (x in xStart to xEnd) {  
  print_int(x);  
  print_string(" ")  
};
```

LOOPS

OTHER ADVANTAGES

- Performance
- Compile Speed
- Tooling: refmt, CLI, Scaffolding, IDEs
- Ecosystem: JS/Node + OCAML
- Language Features: e.g. Keyword Params

TRADE-OFF

- Impurity
- Early Documentation
- No *Async/Await* yet
- Impurity

THREE WISHES





ReasonReact

All your ReactJS knowledge, codified.

WORKSHOP

GET STARTED

```
[@react.component]
let make = (~name) =>
  <button>
    {React.string("Hello!")}
  </button>;
```

Safe and Sound

It's Just Reason. We leverage the existing type system to create a library that types just right.

Playground for Future React

Lightweight, first-class support for the ReactJS community idioms you've been using.

Dro

Easily integrate ReasonReact
a file, quickly

REFERENCES

- Why Reason got started, with Jordan Walke:
<https://reason.town/jordan-interview>
- Sean Grove: The Age of ReasonML
<https://www.youtube.com/watch?v=8LCmLQ1-YqQ>
- Axel Rauschmaier:
<http://reasonmlhub.com/exploring-reasonml>

IMAGE CREDITS

- Jordan Walke at JSConfUS 2013 <https://www.youtube.com/watch?v=GW0rj4sNH2w>
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Thank You!



@MarcoEmrich